

Chapter 1

Change of Coordinates

1.1 Chain Rule in Tensor Notation

The core is: **contract** the notation!

Consider a mapping from independent variables μ, ν to one dependent variable to another.

Using chain rule,

$$\frac{\partial f}{\partial \mu} = \frac{\partial F}{\partial a} \frac{\partial A}{\partial \mu} + \frac{\partial F}{\partial b} \frac{\partial B}{\partial \mu} + \frac{\partial F}{\partial c} \frac{\partial C}{\partial \mu}$$

and

$$\frac{\partial f}{\partial \nu} = \frac{\partial F}{\partial a} \frac{\partial A}{\partial \nu} + \frac{\partial F}{\partial b} \frac{\partial B}{\partial \nu} + \frac{\partial F}{\partial c} \frac{\partial C}{\partial \nu}$$

From here, we use *Einstein Summation Notation* to write them simply.

$$\frac{\partial F}{\partial a^i} \frac{\partial A^i}{\partial \mu}$$

Finally, we also contract μ and ν to μ^i s.t.,

$$\frac{\partial f}{\partial \mu^\alpha} = \frac{\partial F}{\partial a^i} \frac{\partial A^i}{\partial \mu^\alpha} \tag{1.1}$$

1.2 Inverse Functions of Several Variables

Consider the mapping from $\mathbb{Z}$ to Z' and then again to Z . By the chain rule, we have:

$$\frac{\partial Z^i(Z'(Z))}{\partial Z^j} = \frac{\partial Z^i}{\partial Z^{i'}} \frac{\partial Z^{i'}}{\partial Z^j} = J_{i'}^i J_j^{i'} \tag{1.2}$$

Since differentiation of Z^i with respect to Z^j is δ_j^i , we have:

$$J_{i'}^i J_j^{i'} = \delta_j^i \tag{1.3}$$

(1.3) means that $J_{i'}^i$ and $J_j^{i'}$ are inverses of each other. s.t.,

$$(J_{i'})^{-1} = J_j^{i'}$$